Fiscal Year:	FY 2005	**Task Last Updated:**	FY 04/28/2005
PI Name:	Lueptow, Richard M Sc.D.		
Project Title:	Real-Time Gaseous Process Control Monitoring Based on Acoustic Properties		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60208	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-01
Start Date:	07/02/2003	**End Date:**	07/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG9-1544		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The project concerns the development of an acoustic sensor for the reliable monitoring of gas concentrations in on-board environmental control and life-support systems such as the closed-loop oxygen recovery system. In parallel to the development of the actual sensor, theoretical work is in progress aimed at predicting the acoustic attenuation and sound speed in gas mixtures as well as solving the inverse problem of identifying contaminant concentrations. The methodology has been to extend our model of acoustic attenuation to account for hydrogen, oxygen, and carbon dioxide (methane and water vapor are already included in our model). The results of "simulations" of acoustic attenuation under a variety of operating conditions (temperature, pressure, composition) provides information necessary to determine the potential for this approach to process control monitoring or humidity measurement. A small number of experiments have also been performed by an outside acoustics lab to confirm the acoustic attenuation analysis. The objective of this project is to develop a real-time flow-through acoustic sensor to monitor gaseous processes in on-board environmental		

	systems, along with an accurate predictive model. A critical function of life-support systems on a spacecraft is maintaining the air fresh and of nominal composition. Air revitalization involves many situations that will benefit from reliable gaseous process control sensors, such as replenishing of oxygen, removing of excess carbon-dioxide, monitoring air composition, and monitoring humidity levels. Gases that need to be monitored at the nodes of an air revitalization system are oxygen, nitrogen, carbon-dioxide, methane, hydrogen, water vapor, and their mixtures. Another use for reliable sensors is monitoring of the atmospheres of hydroponics chambers, since plant growth depends on nominal gas composition. Acoustic gas sensors are very good candidates for these tasks since they are rugged, virtually calibration- and maintenance-free and give real-time output.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work grew out of earth applications for gaseous fuels, particularly natural gas. Better understanding of acoustic properties of gaseous mixtures permits the potential for the development of real-time gaseous process control monitors or alarm monitors in a wide range of applications including hazardous gas detection, industrial process control, and environmental monitoring. This research will lead to better theoretical models of acoustic properties in gases, thus laying the groundwork for advancing the applications of acoustic sensors. In addition, the methods used to solve the "inverse problem" of extracting the gas composition from acoustic measurements may be applicable in other situations.
Task Progress:	Task 1 – Acoustic Sensor Design Specification: Completed. Task 2 – Gaseous Acoustic Sensor 1 (GAS 1): Completed. The GAS 1 prototype has been developed at Commercial Electronics, our industry partner. Task 3 – Investigation of Acoustic Impedance: Completed. The acoustic impedance of the gas cannot be measured independently of attenuation and sound speed in typical transmit-receive setups. Extra knowledge is needed of either the particle velocity in the sensing volume, measurable with particle velocity sensors or the surface displacement of the piezoelectric transmitter, measurable with either very near-field particle velocity sensors or digital laser Doppler vibrometers. Task 4 – Understanding the “Inverse Problem”: Ongoing. We are working on smart sensing algorithms based on our optimized theoretical models. We have developed an inversion algorithm that is able to infer the nature of contaminant molecules based on operation at only two frequencies. Task 5 – Testing of Gaseous Acoustic Sensor 1: Ongoing. We have been testing the GAS 1 prototype in order to assess its performance in both static and flow conditions. The prototype is able to promptly detect small changes in the speed of sound as a foreign gas leaks in a base gas. In parallel, we are comparing the sensor’s performance to that of a research-grade test cell for a number of gas mixtures. Task 6 – Analysis of Test Data: Ongoing. We have developed programs for data filtering and for obtaining the attenuation and sound speed. We are currently seeking improved algorithms for absolute sound speed measurements. Task 7 – Development of Detection Algorithms: Ongoing. We are fine-tuning our theoretical models of acoustic wave propagation in excitable gases. These predictive tools are in a feed-back loop with experiments: we use measurements to refine the models, while we use the improved models to check measurement consistency. Task 8 – Gaseous Acoustic Sensor 2 (GAS 2): Ongoing. The development of GAS 2 is under way at Commercial Electronics. The second-generation sensor has improved signal processing and conditioning electronics coupled with more efficient built-in programming. It uses on-chip DSP that enables a more robust FFT implementation . In addition, the ultrasonic transducers will have automatic gain control, which eliminates some of the sensitivity issues associated with heavy gases such as CO2. Software for the digital signal processing is also currently under development. Task 9 – Testing of Gaseous Acoustic Sensor 2: Pending. Task 10 – Paper and Report Preparation: Ongoing. Several papers have been written based on the results of this research, as follows (Ed. note: see also Bibliography section): 1) A. G. Petculescu and R. M. Lueptow, “Synthesizing primary molecular relaxation processes in excitable gases using a two-frequency reconstructive algorithm,” in revision for Phys. Rev. Lett, 2005. 2) A. G. Petculescu, R. M. Lueptow, B. Hall, and S. Phillips, “Preliminary results for an ultrasonic gas monitoring system”, Proceedings of the International Conference on Environmental Systems, SAE Paper no 2005-01-2876. 3) A. G. Petculescu and R. M. Lueptow, “Fine-tuning molecular acoustic models: sensitivity of the predicted attenuation to the Lennard-Jones parameters”, J. Acoust. Soc. Am., 117, 175-184 (2005). 4) A. G. Petculescu and R. M. Lueptow, “The impact of the Lennard-Jones parameter choice on theoretical acoustic absorption and dispersion curves”, presented at the 148th Meeting of the Acoustical Society of America, 15-19 November 2004, San Diego, CA.
Bibliography Type:	Description: (Last Updated: 06/28/2021)
Articles in Peer-reviewed Journals	Petculescu AG, Lueptow RM. "Fine-tuning molecular acoustic models: sensitivity of the predicted attenuation to the Lennard–Jones parameters." J Acoust Soc Am. 2005 Jan;117(1):175-84. https://doi.org/10.1121/1.1828547 ; PMID: 15704410 , Jan-2005
Articles in Peer-reviewed Journals	Petculescu AG, Lueptow RM. "Synthesizing primary molecular relaxation processes in excitable gases using a two-frequency reconstructive algorithm." Phys Rev Lett. 2005 Jun 17;94(23):238301. https://doi.org/10.1103/PhysRevLett.94.238301 ; PMID: 16090508 , Jun-2005

Papers from Meeting Proceedings	Petculescu AG, Lueptow RM, Hall B, Phillips S. "Preliminary results for an ultrasonic gas monitoring system." International Conference on Environmental Systems, Rome, Italy, July 2005. International Conference on Environmental Systems, Rome, Italy, July 2005. SAE Technical Paper 2005-01-2876. https://doi.org/10.4271/2005-01-2876 , Jul-2005
Presentation	Lueptow, R. M.; Petculescu, A. G. "Preliminary results for an ultrasonic gas monitoring system" International Conference on Environmental Systems, Rome Jun-2005
Presentation	Petculescu, A. G.; Lueptow, R. M. "The impact of the Lennard-Jones parameter choice on theoretical acoustic absorption and dispersion curves " 148th Meeting of the Acoustical Society of America, San Diego Nov-2004